

1. Adèles, Idèles, and Dirichlet Characters

Let $\chi : (\mathbb{Z} \setminus N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ be a character. Let $\chi : \mathbb{Z} \setminus N\mathbb{Z} \rightarrow \mathbb{C}$ where

$$\chi(a) := \{\chi(a) \text{ if } a \in (\mathbb{Z} \setminus N\mathbb{Z})^\times, 0 \text{ otherwise}\} \quad (1)$$

Definition 1.1 (*Dirichlet Character*) Let $\chi : \mathbb{Z} \setminus N\mathbb{Z} \rightarrow \mathbb{C}$ be defined as above. $\chi_N : \mathbb{Z} \rightarrow \mathbb{C}$ where $n \rightarrow \chi(n \bmod N)$ is a Dirichlet character mod N . Unless otherwise stated χ_A denotes a Dirichlet character mod A .

Suppose χ_N is a Dirichlet character mod N . From χ_N we can get a Dirichlet character χ_M mod M if $N \mid M$. Define $\chi_M := \chi_N \circ \pi_{M,N}$ where $\pi_{M,N} : (\mathbb{Z} \setminus M\mathbb{Z})^\times \rightarrow \mathbb{Z} \setminus N\mathbb{Z}^\times$. Thus we get the following Dirichlet character χ_M :

$$\chi_M(n) := \{\chi_N(n) \text{ if } \gcd(n, M) = 1, 0 \text{ otherwise}\} \quad (2)$$

We can say that χ_M is induced by χ_N . A Dirichlet character χ_N is **primitive** if it isn't induced by any proper divisor $d < N$, and **imprimitive** is when it is induced. The **conductor** of a Dirichlet character χ is the smallest positive integer such that it induces the Dirichlet character χ . The conductor is unique, and it is equal to $N \Leftrightarrow \chi$ is primitive.

$\{(\mathbb{Z} \setminus N\mathbb{Z})^\times, \pi_{M,N}\}$ form a poset, in fact they form an inverse system over $(\mathbb{N}, |)$:

1. $\pi_{N,N} = 1_N$
2. Suppose $N \mid M \mid K$, then $\pi_{K,N} = \pi_{M,N} \circ \pi_{K,M}$

Define,

$$\widehat{\mathbb{Z}}^\times := \varprojlim_N (\mathbb{Z} \setminus N\mathbb{Z})^\times = \left\{ (a_N)_{N \geq 1} \in \prod_{N=1}^{\infty} (\mathbb{Z} \setminus N\mathbb{Z})^\times \mid \text{for all } N \mid M, \pi_{M,N}(a_M) = a_N \right\}. \quad (3)$$

Note $\widehat{\mathbb{Z}}^\times \subset \prod (\mathbb{Z} \setminus N\mathbb{Z})^\times$, and has the subspace topology of the product topology. $\widehat{\mathbb{Z}}^\times$ is a compact (product of compact is compact and closed subset is compact), totally disconnected (each $(\mathbb{Z} \setminus N\mathbb{Z})^\times$ is totally disconnected because discrete and thus the product is totally disconnected) topological group. Such a topology is called **profinite**. Furthermore every Dirichlet character χ_N mod N can be viewed as a *continuous* character from $\widehat{\mathbb{Z}}^\times$ to $\mathbb{C}^\times$ (compose with $\pi_N : \widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z} \setminus N\mathbb{Z})^\times$).

Define for a prime p :

$$\begin{aligned} \mathbb{Z}_p &:= \varprojlim_n \mathbb{Z} \setminus p^n \mathbb{Z} = \left\{ (a_N)_{N \geq 1} \in \prod_{n=1}^{\infty} \mathbb{Z} \setminus p^n \mathbb{Z} \mid \text{for all } N \mid M, \pi_{p^M, p^N}(a_M) = a_N \right\} \\ &= \{\alpha = \alpha_0 + \alpha_1 p + \alpha_2 p^2 + \dots \mid 0 \leq \alpha_i < p\} \end{aligned} \quad (4)$$

1. Inverse Limit $\rightarrow$ Formal Power Series (P-adic expansion): Let $(a_n) \in \mathbb{Z}_p$. We will construct the digits of the p-adic expansion inductively.

Base Case: Let $a_1 = \alpha_0$, viewing a_1 as the unique value in $\{0, \dots, p-1\}$.

Inductive Hypothesis: Assume that for some n , we have α_k for $k < n$ where

$$a_n \equiv \sum_{i=0}^{n-1} \alpha_i p^i \pmod{p^n} \quad (5)$$

Inductive Step: Since $a_{n+1} \equiv a_n \pmod{p^n}$, there is a unique $\alpha_n \in \{0, \dots, p-1\}$ such that,

$$a_{n+1} \equiv \left(\sum_{i=0}^n \alpha_i p^i \right) \bmod p^{n+1}. \quad (6)$$

That is the choice of α_{n+1} .

This produces a unique sequence $(\alpha_n)_{n \geq 0}$ since each choice for α is unique and the formal series is

$$\alpha = \sum_{i=0}^{\infty} \alpha_i p^i \quad (7)$$

2. Formal Power Series $\rightarrow$ Inverse Limits: Let α be a formal power series. Let $(a_n := \alpha \bmod p^n)$.

This is a compatible sequence, and is in the inverse limit.

In fact, ring structure is preserved as well.

$$\mathbb{Z}_p^\times \cong \{ \alpha \in \mathbb{Z}_p \mid \alpha \bmod p \neq 0 \} \quad (8)$$

Note that

$$\widehat{\mathbb{Z}}^\times \cong \prod_{p \text{ prime}} \mathbb{Z}_p^\times \quad (9)$$

because the isomorphism induced by chinese remainder theorem commutes with the maps in the inverse system.

We also have the p-adic numbers:

$$\mathbb{Q}_p := \left\{ \alpha = \sum_{i=v}^{\infty} a_i p^i \mid 0 \leq a_i < p, v \in \mathbb{Z} \right\} \quad (10)$$

Note the p-adic valuation is the index of the first nonzero element of the formal Laurent Series. Note $\mathbb{Q} \subset \mathbb{Q}_p$ densely, similarly $\mathbb{Z} \subset \mathbb{Z}_p$.

Definition 1.2 P-adic valuation induces an absolute value on $\mathbb{Q}$, where $\forall n \in \mathbb{Q}$ let $v_p(n)$ be the p-adic valuation of n . Then

$$|v|_p = p^{-v_p(n)}. \quad (11)$$

Definition 1.3 Let $|\cdot|_\infty$ be the standard absolute value on $\mathbb{Q}$.

These absolute values give a different notion of closeness.

Example 1.4 Let $n = 1000 = 8 * 125 = 2^3 * 5^3$.

1. $|n|_\infty = 1000$ so n is pretty far from 0.
2. $|n|_2 = 2^{-3} = \frac{1}{8}$ so n is close to 0 in 2-adic absolute value.
2. $|n|_5 = 5^{-3} = \frac{1}{125}$ so n is even closer to 0 in 5-adic absolute value.

The different absolute values give a different closure. The closure of $\mathbb{Q}$ with respect to $|\cdot|_\infty$ is $\mathbb{R}$.

The closure of $\mathbb{Q}$ with respect to $|\cdot|_p$ is $\mathbb{Q}_p$.

Definition 1.5 For a field K (in our context $\mathbb{Q}$). Let A be the set of absolute values $|\cdot| : K \rightarrow \mathbb{R}_{\geq 0}$.

Define the equivalence relation $\sim$ on A where two absolute values are equivalent if they generate the same topology. The **places** of A is $\sim / A$. A **place** of K is the equivalence class of 1 absolute value.

By Ostrowski's theorem, the places of $\mathbb{Q}$ are $\{|\cdot|_\infty\} \cup \{|\cdot|_p : p \text{ prime}\}$. A key idea in number theory, is that one can deduce global properties of $\mathbb{Q}$ from local properties of $\mathbb{Q}$, which come from properties of its completion with respect to its various places. This principle is known as the local-to-global principle or Hasse's Principle. One possible approach to this is to combine all the local information so we have a mathematical object that contains all the local information in a way that illuminates the broad behavior of $\mathbb{Q}$ while still respecting $\mathbb{Q}$'s global behavior. The naive way is to construct

$$\mathbb{A}' = \mathbb{R} \times \prod_{p \text{ prime}} \mathbb{Q}_p, \quad (12)$$

but this set is too unwieldy an element in the set $\alpha = (\alpha_\infty, \alpha_2, \alpha_3, \alpha_5, \dots)$ is a product of completely independent components.

1. You can trivially embed $\mathbb{Q} \subset \mathbb{A}'$ where $\alpha \rightarrow (\alpha, \alpha, \dots)$, but this subgroup of $\mathbb{A}'$ has no special or useful properties within the vastness of the full product. The product is so large that the relationships between rational numbers become trivial.
2. The topology also kinda sucks. For any useful tools of analysis to be applied you need to be locally compact. The direct product is not locally compact and thus becomes unworkable "wasteland." All the powerful machinery you want to use breaks down.

Okay so what do we do. For any $r \in \mathbb{Q}$, $r = \frac{p}{q}$ where $\gcd(p, q) = 1$. $q = s_1^{m_1} \dots s_n^{m_n}$ for some n . So for all primes t except $s_1, \dots, s_n$, $|q|_t = 1$, thus $d \in \mathbb{Z}_t$. Thus for all rationals, they are in $\mathbb{Z}_p$ for all but finitely many primes. We want to capture this structure in our ring, so we can just enforce that.

$$\mathbb{A} = \mathbb{R} \times \prod_{p \text{ prime}}^* (\mathbb{Q}_p) \quad (13)$$

Definition 1.6 $\mathbb{A}$ is the adèle ring of $\mathbb{Q}$. The $\prod_{p \text{ prime}}^*$ denotes the restricted product, where $\forall (a_n) \in \mathbb{A}$ all but finitely many components of (a_n) must lie in $\mathbb{Z}_p$. Let

$$\mathbb{A}_f := \prod_{p \text{ prime}}^* (\mathbb{Q}_p), \quad (14)$$

thus $\mathbb{A} = \mathbb{R} \times \mathbb{A}_f$.

This gives us the needed structure for $\mathbb{Q}$. Now lets check the topology. The adèle ring is given the restricted product topology. All basis sets of $\mathbb{A}$, look like

$$U_\infty \times \prod_{p \in S} U_p \times \prod_{p \notin S} \mathbb{Z}_p \quad (15)$$

where S is a finite set of primes, $U_\infty \subset \mathbb{R}$ open, and $U_p \subset \mathbb{Q}_p$ open. $\mathbb{Z}_p$ is compact, and thus the product is compact. By combining with finite product of locally compact spaces $\mathbb{R}$ and $U_p \subset \mathbb{Q}_p$, the basis set is locally compact. Thus $\mathbb{A}$ is locally compact. So it has the necessary topological data. Furthermore, $\mathbb{Q} \hookrightarrow \mathbb{A}$, where $\alpha \rightarrow (\alpha, \dots)$, but now its a discrete subgroup of $\mathbb{A}$.

Definition 1.7 The idele group of $\mathbb{Q}$ is the restricted product

$$\mathbb{A}^\times = \prod_{p \text{ prime}}^* \mathbb{Q}_p^\times \quad (16)$$

with respect to $\mathbb{Z}_p^\times$. Again there is the natural diagonal embedding of $\mathbb{Q}^\times \hookrightarrow \mathbb{A}^\times$.

Theorem 1.8 (Decomposition of Adele Ring)

$$\mathbb{A}^\times \cong \mathbb{Q}^\times \times \mathbb{R}_+^\times \times \widehat{\mathbb{Z}}^\times \quad (17)$$

under this decomposition let $x = \alpha t u$ for $\alpha \in \mathbb{Q}^\times$, $t \in \mathbb{R}_+^\times$, and, $u \in \widehat{\mathbb{Z}}^\times$.

Proof $\forall \alpha \in \mathbb{A}^\times$, $\alpha = (\alpha_\infty, \alpha_2, \alpha_3, \dots)$. $\alpha_\infty \in \mathbb{R}^\times$ and $\alpha_\infty = \text{sgn}(\alpha_\infty)|\alpha_\infty|$. Thus

$$\mathbb{R}^\times \cong \mathbb{R}_{>0}^\times \times \{\pm 1\} \quad (18)$$

. For each prime p , $\exists! (u_p, k_p) \in \mathbb{Z}_p^\times \times \mathbb{Z}$ such that $\alpha_p := p^{k_p} u_p$. Thus

$$\mathbb{Q}_p \cong p^\mathbb{Z} \times \mathbb{Z}_p^\times \quad (19)$$

But note that for α for all but finitely many primes $\alpha_p \in \mathbb{Z}_p^\times$. Thus only finitely many $k_p \neq 0$. Thus the sequence $(p^{k_p})_{p \text{ prime}}$ is in

$$\bigoplus_{i=0}^{\infty} p^\mathbb{Z} = \left\{ (p^{k_p}) \in \prod_{i=0}^{\infty} \mathbb{Z} \mid k_p = 0 \text{ for almost all } p \right\} \quad (20)$$

Thus

$$\begin{aligned} \prod_{p \text{ prime}}^* \mathbb{Q}_p &\cong \prod_{p \text{ prime}}^* \mathbb{Z} \times \mathbb{Z}_p^\times \\ &\cong \left(\prod_{p \text{ prime}} \mathbb{Z}_p^\times \right) \times \bigoplus_{p \text{ prime}} p^\mathbb{Z} \\ &\cong \widehat{\mathbb{Z}}^\times \times \bigoplus_{p \text{ prime}} p^\mathbb{Z} \end{aligned} \quad (21)$$

Thus

$$\begin{aligned} \mathbb{A}^\times &\cong \{\pm 1\} \times \mathbb{R}_+^\times \times \widehat{\mathbb{Z}}^\times \times \bigoplus_{p \text{ prime}} p^\mathbb{Z} \\ &= \left(\{\pm 1\} \times \bigoplus_{p \text{ prime}} p^\mathbb{Z} \right) \times \mathbb{R}_+^\times \times \widehat{\mathbb{Z}}^\times \\ &= \mathbb{R}_+^\times \times \mathbb{Q}^\times \times \widehat{\mathbb{Z}}^\times \end{aligned} \quad (22)$$

□

From this decomposition we see that any Dirichlet Charatcer χ defines a continous character

$$\omega : \mathbb{Q}^\times / \mathbb{A}^\times \rightarrow \mathbb{C}^\times \quad (23)$$

where $\omega(x) = \omega(\alpha t u)$ applying the decomposition equivalence and $\omega(\alpha t u) = \chi(u)$.

Definition 1.9 $\forall x \in \mathbb{A}$, $x = \alpha t u$, $|\cdot| : \mathbb{A} \rightarrow \mathbb{R}_{>0}$ where $x = \alpha t u \rightarrow |t|$ is a absolute value on $\mathbb{A}$.

Definition 1.10 A **quasicharacter** of a group G is a homomorphism from G to $\mathbb{C}^\times$.

Any quasicharacter of $\mathbb{A}$ is of the form $\omega|\cdot|^s$ for some $s \in \mathbb{C}$. The Dirichlet characters are quasicharacters of finite order.

Okay now here is the kicker: By Chinese Remainder theorem,

$$(\mathbb{Z} \setminus N\mathbb{Z})^\times = \prod_{p \text{ prime}} ((\mathbb{Z} \setminus p^{n_p}\mathbb{Z})^\times), \quad (24)$$

any Dirichlet character, χ_N has the factorization

$$\chi_N = \otimes_p (\chi_N)_p \quad (25)$$

where $(\chi_N)_p$ is the induced Dirichlet in $(\mathbb{Z} \setminus p^n \mathbb{Z})^\times$. This gives rise to the factorization, $\chi = \otimes_p \chi_p$ of the corresponding character χ of $\widehat{\mathbb{Z}}^\times$. Note that for each place v of $\mathbb{Q}$, there is a natural inclusion $\mathbb{Q}_v \hookrightarrow \mathbb{A}^x \setminus \mathbb{Q}^x$, $x_v \rightarrow (\dots, 1, 1, x_v, 1, 1, \dots)$. Thus any quasicharacter ω determines quasicharacters ω_v of each $\mathbb{Q}_v^\times$. Thus there is a factorization $\omega = \otimes_v \omega_v$.

Theorem 1.11 (*Strong Approximation Theorem for $\mathbb{Q}$*) Let S to be finite set of primes. Let $f : S \rightarrow \mathbb{Q}$ where $s \rightarrow a_s$. $\forall \varepsilon > 0$, $\exists a \in \mathbb{Q}$ such that $|a - a_s|_s < \varepsilon$ and $|a|_p \leq 1$ ie $a \in \mathbb{Z}_p$ for all $p \notin S$.

Proof Let S and f be defined as above. $\forall a_s, a_s = \frac{b_s}{c_s}, b_s, c_s \in \mathbb{Z}$ such that $\gcd(b_s, c_s) = 1$. The idea, to ensure $|a|_p \leq 1$ for all $p \notin S$, a has to be of the form $\frac{x}{D}$ (wlog $\gcd(x, D) = 1$) where $D = \prod_{s \in S} s^{K_s}$. $\forall s \in S$, let

$$K_s := \max(v_p(c_p) : p \in S) \quad (26)$$

where v_s is the s -adic valuation. $\forall s \in S, v_s(Da) = v_s(D) + v_s(a_p) = v_s(D) + v_s(b_s) - v_s(c_s) \geq v_s(b_s) \geq 0$. Let $b_s' = Da_s$ which is now a s -adic integer. Thus $\forall z \in \mathbb{Z}, z - b_s'$ is a s -adic integer and $|z - b_s'|_s \leq 1$, thus $\exists N_s \in \mathbb{N}$ such that $\forall n \geq N_s$

$$\begin{aligned} z - b_s' &\equiv 0 \pmod{s^n} \\ x &\equiv b_s' \pmod{s^n} \end{aligned} \quad (27)$$

$\forall (n_s)_{s \in S} \in \mathbb{N}^S \ni n_s \geq N_s$. By Chinese Remainder Theorem, $\exists x$ solution to

$$\begin{aligned} x &\equiv b_{s_1}' \pmod{s_1^{n_{s_1}}} \\ &\vdots \\ x &\equiv b_{s_{|S|}}' \pmod{s_{|S|}^{n_{|S|}}} \end{aligned} \quad (28)$$

Thus, $|x - a_s|_s = s^{-n_s} \leq s^{-N_s}$.

$\forall \varepsilon > 0, \forall s \in S \exists N_s \in \mathbb{N} \ni s^{-N_s} < \varepsilon$ $|D|_s = \varepsilon s^{-K_s}$. Take $N = \max(N_s \mid s \in S)$. Let x be the solution to the above system with the tuple $(N, \dots, N)$. Let $a = \frac{x}{D}$. Thus,

$$|a - a_s|_s = \frac{|x - Da_s|_s}{|D|_s} = \frac{|x - b_s'|_s}{|D|_s} \leq \frac{s^{-N}}{|D|_s} < \varepsilon \quad (29)$$

Additionally, since D is only divisible by primes in S , $\forall p \notin S, v_p(D) = 0$. Thus $v_p(a) = v_p(x) - v_p(D) = v_p(x) \geq 0$. Thus $|a|_p \leq 1$. $\square$

Corollary 1.12 (*Strong Approximation Theorem for Adeles*) The image of the diagonal embedding $\mathbb{Q} \hookrightarrow \mathbb{A}_f (q \rightarrow (q, q, \dots))$ is dense.

Proof To show that the $\mathbb{Q}$ is dense under the diagonal embedding we have to show that it intersects with every open set. It suffices to show this for all basis open sets U . As described above, $\forall U$ a element of the basis of the topology of $\mathbb{A} \exists S \subset \mathbb{P}$ finite such that

$$U = \left(\prod_{p \in S} B(a_p, \varepsilon) \right) \times \left(\prod_{p \in \mathbb{P} - S} \mathbb{Z}_p \right). \quad (30)$$

This is slightly different that what it is above (for each $s \in S$ we had opens $U_s \subset \mathbb{Q}_s$ but we can just take ε_s balls as they are basis sets of the topology of $\mathbb{Q}_s$ and then just use the same ε for all s , we

also had a open set from $\mathbb{R}$ which we don't need anymore because we are looking at $\mathbb{A}_f$). By Theorem 1.11, $\exists q \in \mathbb{Q}$ such that $|q|_s$ and $|q|_p \leq 1$ ($q \in \mathbb{Z}_p$) for all $(s, p) \in (S, \mathbb{P} - S)$. Thus $q \in U \cap \mathbb{Q}$. Thus $\mathbb{Q}$ is dense in $\mathbb{A}_f$, you can extend this to $\mathbb{Q}^\times \subset \mathbb{A}^\times$ under the same diagonal map. $\square$

Why is it natural to factor through $\mathbb{Q}^\times / \mathbb{A}^\times$ rather than say $\mathbb{A}^\times$ or $(\mathbb{Q}^\times \times \mathbb{R}_{>0}^\times) / \mathbb{A}^\times$?

The adèle ring $\mathbb{A}$ and $\mathbb{A}^\times$ hold all local information of $\mathbb{Q}$, with all its completions simultaneously. However as shown $\mathbb{Q} \hookrightarrow \mathbb{A}$ and $\mathbb{Q}^\times \hookrightarrow \mathbb{A}^\times$ naturally with the diagonal map which acts in a “global” way. When we form the quotient $\mathbb{Q}^\times / \mathbb{A}^\times$, we are essentially saying: “Let’s consider all the local data, but let’s identify any two ideles if they differ only by multiplication by a global number.” This procedure filters out the “trivial” global structure to isolate the interactions and relationships between the different local components that are not already explained by the global field $\mathbb{Q}$ itself. Thus factoring through $\mathbb{A}$ doesn’t make sense because the arithmetic behavior becomes obscured. Now what about, $(\mathbb{Q}^\times \times \mathbb{R}_{>0}^\times) / \mathbb{A}^\times$. While Dirichlet characters are finite order and act trivially on $\mathbb{R}_{>0}^\times$ not all quasicharacters are finite order and thus act nontrivially on $\mathbb{R}_{>0}^\times$ as shown above.

2. From Modular Forms to Automorphic Representations

Much like the correspondence between χ_N Dirichlet characters mod N and quasicharacters $\omega : \mathbb{Q}^\times / \mathbb{A}^\times \rightarrow \mathbb{C}^\times$, there exists a very similar correspondence as we increase dimensions. Here is some stuff which carries over:

1. There is a factorization of characters: For a automorphic representation π of $\mathrm{GL}_2(\mathbb{A})$, π will have the factorization

$$\pi = \otimes_v \pi_v \quad (31)$$

where v is over the places of $\mathbb{Q}$.

2. Furthermore, just like there are many Dirichlet characters for each quasicharacter, there is a distinguished one which is the primitive Dirichlet character, a similar situation happens in $\mathrm{GL}_2(\mathbb{A})$. There are infinitely many modular forms f corresponding to a automorphic representation of $\mathrm{GL}_2(\mathbb{A})$ but there is a unique distinguished primitive form, called a new form.

Okay so what are the differences, the representation π and its local components (factors) π_v may be infinite dimensional so the representation is much more complicated. Furthermore π can be realized in the “action” of $\mathrm{GL}_2(\mathbb{A})$ by right translations of functions of $\mathrm{GL}_2(\mathbb{Q}) / \mathrm{GL}_2(\mathbb{A})$.

Definition 2.1 For $k > 2$ even integer. Let $f \in S_k(\Gamma_0(N), \chi_N)$ where χ_N is a Dirichlet character mod N . Thus f is a holomorphic cusp for where $\forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid c \equiv 0 \pmod{N} \right\}$ and $\forall z \in \mathbb{H}$,

$$f\left(\frac{az+b}{cz+d}\right) = \chi(d)(cz+d)^k f(z) \quad (32)$$

and the Fourier expansion of f is

$$f(x) := \sum_{n=1}^{\infty} a_n e(nz). \quad (33)$$

Definition 2.2 Let G be the function of the space of fields to the space of all groups of invertible matrices of 2 dimensions such that $F \rightarrow \mathrm{GL}_2(F)$. Thus $G(\mathbb{A}) = \mathrm{GL}_2(\mathbb{A})$. Let G_v be defined as follows:

$$G_v := \begin{cases} \mathrm{GL}_2(\mathbb{R}) & \text{if } v = \infty \\ \mathrm{GL}_2(\mathbb{Q}_p) & \text{if } \mathbb{Q}_v = \mathbb{Q}_p \end{cases} \quad (34)$$

$$G(\mathbb{A}) := G(\mathbb{R} \times \mathbb{A}_f) \cong G(\mathbb{R}) \times G\left(\prod_{p \in \mathbb{P}}^* \mathbb{Q}_p\right) = G(\mathbb{R}) \times \prod_{p \in \mathbb{P}}^* G(\mathbb{Q}_p) \quad (35)$$

where the restricted product is taken with respect to the subgroups $K_p := G(\mathbb{Z}_p) \subset G(\mathbb{Q}_p)$. $G(\mathbb{R})$ and $G(\mathbb{A}_f)$ is a locally compact topological group and

$$K = \prod_{p \in \mathbb{P}} K_p \leq G(\mathbb{A}_f) \quad (36)$$

is a compact open subgroup. Finally just like the 1d case, $G(\mathbb{Q}) \hookrightarrow G(\mathbb{A})$ where $\gamma \rightarrow (\gamma, \gamma, \dots)$ under the diagonal embedding, furthermore under the inclusion $G(\mathbb{Q}) < G(\mathbb{A})$ is a discrete subgroup. We get a similar construction with SL .

Lemma 2.3 Suppose $\forall S \subset \mathbb{P}, \forall s \in S$ we are given $g_s \in \mathrm{SL}_2(\mathbb{Q}_s)$. There $\exists \gamma \in \mathrm{SL}_2(\mathbb{Q})$ such that:

1. All the matrix entries of γ lie in $\mathbb{Z}_p$ for $p \notin S$
2. For each $p \in S$, $\gamma^{-1}g_p \in \mathrm{SL}_2(\mathbb{Z}_p)$

Proof Base Case: Suppose $S = \{p\}$. Let $\mathbb{Q}_{(p)}$ be the ring of rationals with powers of p in the denominator. For $q \neq p$, $\mathbb{Q}_{(p)} \subset \mathbb{Q}_q$ also lies in $\mathbb{Z}_q$. Thus if we find a $\gamma \in \mathrm{SL}_2(\mathbb{Q}_{(p)})$ that satisfies condition 2, we automatically satisfy condition 1 as well. Let $g = g_p$. By the Elementary divisors theorem for $M_2(\mathbb{Q}_p)$, $\exists \delta_1, \delta_2$ such that

$$g = \delta_1 d \delta_2 \quad (37)$$

where $d = \mathrm{diag}(p^m, p^{-m})$ for some $m \geq 0$. By that $\mathrm{SL}_2(\mathbb{Z}) \twoheadrightarrow \mathrm{SL}_2(\mathbb{Z} \setminus p\mathbb{Z})$, $\exists \gamma_i$ such that

$$\gamma_i \equiv \delta_i \pmod{p^{2m}}. \quad (38)$$

Let $\gamma = \gamma_1 d \gamma_2$.

Then

$$\gamma^{-1}d = (\gamma_1 d \gamma_2)^{-1}(\delta_1 d \delta_2) \equiv I \pmod{p^{2m}}. \quad (39)$$

Thus $\gamma^{-1}g \in \mathrm{SL}_2(\mathbb{Z}_p)$.

Inductive Hypothesis: Assume for some n that $\forall S \subset \mathbb{P} \ni |S| = n$ and for all choices of g_p for $p \in S$, that $\exists \gamma \in \mathrm{SL}_2(\mathbb{Q})$ that the conditions of the theorem hold.

Inductive Step: Let $S = \{p\} \cup \{p_1, \dots, p_n\} \subset \mathbb{P}$ and let $p \rightarrow g$ and $p_i \rightarrow g_i$ where $g \in \mathrm{SL}_2(\mathbb{Q}_p)$ and $g_i \in \mathrm{SL}_2(\mathbb{Q}_{p_i})$. By the base case, $\exists \gamma_1$ such that $\gamma_1 \in \mathrm{SL}_2(\mathbb{Z}_q)$ for $q \neq p$. By the inductive hypothesis $\exists \gamma_2$ for $S = \{p_1, \dots, p_n\}$ and $\{\gamma_1^{-1}g_1, \dots, \gamma_1^{-1}g_n\}$. Let $\gamma = \gamma_1 \gamma_2$, thus $\gamma \in \mathrm{SL}_2(\mathbb{Z}_q)$ for all $q \notin S$. Furthermore,

$$\begin{aligned} \gamma^{-1}g &= \gamma_2^{-1}\gamma_1^{-1}g \in \mathrm{SL}_2(\mathbb{Z}_p) \\ \gamma^{-1}g_i &= \gamma_2^{-1}\gamma_1^{-1}g_i \in \mathrm{SL}_2(\mathbb{Z}_{p_i}) \end{aligned} \quad (40)$$

Thus this holds for all finite sets S and all choices of g_p for $p \in S$. □

Theorem 2.4 (Strong Approximation for $\mathrm{SL}_2(\mathbb{Q})$) Suppose $\forall S \subset \mathbb{P}, \forall s \in S$ we are given $g_s \in \mathrm{SL}_2(\mathbb{Q}_s)$ and $n_p \in \mathbb{N}_0$. There $\exists \gamma \in \mathrm{SL}_2(\mathbb{Q})$ such that:

1. All the matrix entries of γ lie in $\mathbb{Z}_p$ for $p \notin S$
2. For each $p \in S$, $\gamma^{-1}p \in \text{SL}_2(\mathbb{Z}_p)$ and $\gamma^{-1}g_p \equiv I \pmod{p^{n_p}}$

Proof By Lemma 2.3, $\exists \eta \in \text{SL}_2(\mathbb{Q})$ such that:

1. All the matrix entries of η lie in $\mathbb{Z}_p$ for $p \notin S$
2. For each $p \in S$, $\eta^{-1}g_p \in \text{SL}_2(\mathbb{Z}_p)$

Let $N = \prod_{p \in S} p^{n_p}$. By Chinese Remainder Theorem,

$$\text{SL}_2(\mathbb{Z} \setminus N\mathbb{Z}) \cong \prod_{p \in S} \text{SL}_2(\mathbb{Z} \setminus p\mathbb{Z}). \quad (41)$$

Thus $\exists \bar{h} \in \text{SL}_2(\mathbb{Z} \setminus N\mathbb{Z})$ such that

$$\bar{h} \equiv \eta^{-1}g_p \pmod{p^{n_p}} \quad (42)$$

for all $p \in S$. Furthermore since the quotient map $\text{SL}_2(\mathbb{Z}) \twoheadrightarrow \text{SL}_2(\mathbb{Z} \setminus N\mathbb{Z})$ is surjective, then $\exists h \in \text{SL}_2(\mathbb{Z})$ such that $h \twoheadrightarrow \bar{h}$ and satisfies all the same congruences. Let $\gamma = \eta h$ and since h is a integral matrix $\gamma \in \text{SL}_2(\mathbb{Z}_p)$ for all $p \notin S$. $\forall p \in S$,

$$\gamma^{-1}g_p = (\eta h)^{-1}g_p = h^{-1}\eta^{-1}g_p \equiv g_p^{-1}\eta\eta^{-1}g_p \pmod{p^{n_p}} = I \pmod{p^{n_p}}. \quad (43)$$

Thus γ is the needed member of $\text{SL}_2(\mathbb{Q})$. □

Theorem 2.5 Suppose that for each prime p we have a compact open $K_p \subset \text{SL}_2(\mathbb{Q}_p)$ where for almost all primes $K_p = \text{SL}_2(\mathbb{Z}_p)$. Let $K_f = \prod_{p \in \mathbb{P}} K_p$. Then

$$\text{SL}_2(\mathbb{A}) = \text{SL}_2(\mathbb{Q}) \cdot (\text{SL}_2(\mathbb{R}) \times K_f) \quad (44)$$

or equivalently $\text{SL}_2(\mathbb{Q}) \times \text{SL}_2(\mathbb{R})$ is dense in $\text{SL}_2(\mathbb{A})$.

Proof $\forall g = \left(g_\infty, (g_p)_{p \in \mathbb{P}} \right) \in \text{SL}_2(\mathbb{A})$. We want to find a $\gamma \in \text{SL}_2(\mathbb{Q})$ such that $\gamma^{-1}g \in \text{SL}_2(\mathbb{R}) \times K_f$. We have 2 conditions (1 is trivial):

1. $\gamma^{-1}g_\infty \in \text{SL}_2(\mathbb{R})$. This is always true since $\text{SL}_2(\mathbb{Q}) \subset \text{SL}_2(\mathbb{R})$.
2. $(\gamma^{-1}g)_f = (\gamma^{-1}g_p)_p \in K_f$ so $\gamma^{-1}g_p \in K_p$ for all primes.

Let $S = \{p \in \mathbb{P} \mid K_p \neq \text{SL}_2(\mathbb{Z})\} \cup \{p \in \mathbb{P} \mid g_p \notin K_p\}$, both sets are finite, thus S is finite. For any compact open subgroup K of $\text{SL}_2(\mathbb{Q}_p)$, $\exists n_p \in \mathbb{N}$ such that

$$\{A \in \text{SL}_2(\mathbb{Z}_p) \mid A \equiv I \pmod{p^{n_p}}\} \subset K. \quad (45)$$

Thus for each $p \in S$, $\exists n_p \geq 1$. By Theorem 2.4, $\exists \gamma \in \text{SL}_2(\mathbb{Q})$ such that:

1. $\gamma \in \text{SL}_2(\mathbb{Z}_q)$ for $q \notin S$.
2. $\gamma^{-1}g_p \in \text{SL}_2(\mathbb{Z}_p)$ and $\gamma^{-1}g_p \equiv I \pmod{p^{n_p}}$ for $p \in S$.

For $p \in S$, the second condition on γ tells us that $\gamma^{-1}g_p \in K_p$. For $p \notin S$ then $p \notin \{p \in \mathbb{P} \mid K_p \neq \text{SL}_2(\mathbb{Z})\} \Rightarrow K_p = \text{SL}_2(\mathbb{Z})$ and since $p \notin \{p \in \mathbb{P} \mid g_p \notin K_p\}$ that means $g_p \in K_p = \text{SL}_2(\mathbb{Z})$. By the first condition on γ , we know $\gamma \in \text{SL}_2(\mathbb{Z}_p)$. Thus $\gamma_p^{-1}g_p \in K_p = \text{SL}_2(\mathbb{Z})$. Thus

$$\gamma^{-1}g \in \text{SL}_2(\mathbb{R}) \times K_f \Rightarrow g \in \gamma \cdot (\text{SL}_2(\mathbb{R}) \times K_f). \quad (46)$$

Thus

$$\mathrm{SL}_2(\mathbb{A}) = \mathrm{SL}_2(\mathbb{Q}) \cdot (\mathrm{SL}_2(\mathbb{R}) \times K_f). \quad (47)$$

□

Corollary 2.6 Let K_f be a compact open subgroup of $\mathbb{A}_f$. Let $\Gamma_{K_f} = \mathrm{SL}_2(\mathbb{Q}) \cap (\mathrm{SL}_2(\mathbb{R}) \times K_f)$ where $\mathrm{SL}_2(\mathbb{Q})$ is viewed as a diagonal embedding into $\mathrm{SL}_2(\mathbb{A})$. Then

$$\Gamma_{K_f} \backslash \mathrm{SL}_2(\mathbb{R}) \leftrightarrow \mathrm{SL}_2(\mathbb{Q}) \backslash \mathrm{SL}_2(\mathbb{A}) / K_f. \quad (48)$$

Proof Note,

$$\Gamma_{K_f} \backslash \mathrm{SL}_2(\mathbb{R}) \leftrightarrow \Gamma_{K_f} \backslash \mathrm{SL}_2(\mathbb{R}) \times K_f / K_f \leftrightarrow \mathrm{SL}_2(\mathbb{Q}) \backslash \mathrm{SL}_2(\mathbb{Q}) \cdot (\mathrm{SL}_2(\mathbb{R}) \times K_f) / K_f \quad (49)$$

Where the last bijection comes from the fact that $\forall H, K \leq G, H \cap K \backslash K \leftrightarrow H \backslash H \cdot K$. Finally by Theorem 2.5,

$$\mathrm{SL}_2(\mathbb{Q}) \backslash \mathrm{SL}_2(\mathbb{Q}) \cdot (\mathrm{SL}_2(\mathbb{R}) \times K_f) / K_f \leftrightarrow \mathrm{SL}_2(\mathbb{Q}) \backslash \mathrm{SL}_2(\mathbb{A}) / K_f. \quad (50)$$

□

Corollary 2.7 Suppose that $H \subset G(\mathbb{A}_f)$ is a compact open subgroup such that $\det(H) = \widehat{\mathbb{Z}}^\times$. Then

$$G(\mathbb{A}) = G(\mathbb{Q}) \times G(\mathbb{R})^+ \times H \quad (51)$$

and

$$G(\mathbb{Q}) \backslash G(\mathbb{A}) / H \cong \Gamma \backslash G(\mathbb{R})^+ \quad (52)$$

where $G(\mathbb{R})^+ := \{g \in G(\mathbb{R}) \mid \det(g) > 0\}$ and $\Gamma = G(\mathbb{Q}) \cap (G(\mathbb{R}^+) \times H) \leq G(\mathbb{R})^+$ via projection into the archimedean component.

Proof Note $\det : G(\mathbb{A}) \twoheadrightarrow \mathbb{A}^\times \cong \mathbb{Q}^\times \mathbb{R}_+^\times \widehat{\mathbb{Z}}^\times$. We will match the determinants, $\det(G(\mathbb{Q})) = \mathbb{Q}^\times$. By the assumption on H , $\det(G(\mathbb{R})^+ \times H) = \mathbb{R}_+^\times \times \mathbb{Q}^\times$. Take $g \in G(\mathbb{A})$, by the decomposition of $\mathbb{A}^\times$, $\det(g) = \alpha\beta$ for $\alpha \in \mathbb{Q}^\times$, $\beta \in \mathbb{R}_+^\times \times \widehat{\mathbb{Z}}^\times$. Let $q \in \det^{-1}(\alpha)$ and $x \in \det^{-1}(\beta)$. Let $g' = q^{-1}gx^{-1}$. $\det(g') = \det(q)^{-1} \det(g) \det(x)^{-1} = \alpha^{-1} \alpha \beta \beta^{-1} = 1$. Thus $g' \in \mathrm{SL}_2(\mathbb{A})$. By Theorem 2.5, $g' = \gamma y$ for $\gamma \in \mathrm{SL}_2(\mathbb{Q})$ and $y \in (\mathrm{SL}_2(\mathbb{R}) \times K_f)$ where $K_f = H \cap \mathrm{SL}_2(\mathbb{A}_f)$. Thus $y \in \mathrm{SL}_2(\mathbb{R}) \times (H \cap \mathrm{SL}_2(\mathbb{A}_f))$. Thus $g = qg'x = q\gamma yx$, $q, \gamma \in G(\mathbb{Q}) \Rightarrow q\gamma \in G(\mathbb{Q})$ and $y, x \in G(\mathbb{R})^+ \times H \Rightarrow yx \in G(\mathbb{R})^+ \times H$. Thus

$$G(\mathbb{A}) = G(\mathbb{Q}) \cdot (G(\mathbb{R})^+ \times H) \quad (53)$$

.

By almost exactly the same argument as Corollary 2.6, the statement holds for the double coset. □

Example 2.8 Take

$$H = K = \prod_{p \in \mathbb{P}}^* K_p. \quad (54)$$

where $K_p = \mathrm{GL}_2(\mathbb{Z}_p)$. Then

$$\begin{aligned}
\Gamma &= G(\mathbb{Q}) \cap (G(\mathbb{R})^+ \times H) \cong G(\mathbb{Q}) \cap \left(G(\mathbb{R})^+ \times \prod_{p \in \mathbb{P}}^* \mathrm{GL}_2(\mathbb{Z}_p) \right) \\
&= G(\mathbb{Q})^+ \times \left(G(\mathbb{Q})^+ \cap \prod_{p \in \mathbb{P}}^* \mathrm{GL}_2(\mathbb{Z}_p) \right)
\end{aligned} \tag{55}$$

$\forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(\mathbb{Q})^+ \cap \mathrm{GL}_2(\mathbb{Z}_2) \cap \mathrm{GL}_2(\mathbb{Z}_3) \cap \dots$ that means, $a, b, c, d \in \mathbb{Z}_p$ for all $p \in \mathbb{P}$. Additionally it means $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \in \mathbb{Z}_p^\times$ for all p . That implies $ad - bc = \pm 1$ and since the determinant is positive, that means $ad - bc = 1$.

Thus

$$\Gamma = \mathrm{SL}_2(\mathbb{Z}) \times \prod_{p \in \mathbb{P}} \mathrm{SL}_2(\mathbb{Z}) \cong \mathrm{SL}_2(\mathbb{Z}), \tag{56}$$

again passing through the diagonal embedding.

Example 2.9 Take

$$H = K_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(\mathbb{Z}) \mid c \equiv 0 \pmod{N} \right\}. \tag{57}$$

Note $H \in G(\mathbb{Z}_p)$ for all $p \in \mathbb{P}$.

$$\begin{aligned}
\Gamma &= G(\mathbb{Q}) \cap \left(G(\mathbb{R})^+ \times \prod_{p \in \mathbb{P}} H \right) \\
&= G(\mathbb{Q})^+ \times \prod_{p \in \mathbb{P}} (G(\mathbb{Q})^+ \cap H) \\
&= G(\mathbb{Z})^+ \times \prod_{p \in \mathbb{P}} (G(\mathbb{Z}) \cap H) \\
&= \Gamma_0(N) \times \prod_{p \in \mathbb{P}} \Gamma_0(N) \cong \Gamma_0(N)
\end{aligned} \tag{58}$$

Definition 2.10 There exists a surjective homomorphism from $K_0(N) \twoheadrightarrow (\mathbb{Z} \setminus N\mathbb{Z})^\times$ where

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \twoheadrightarrow a \pmod{N} \tag{59}$$

so a Dirichlet character χ_N defines a character χ of $K_0(N)$.

$G(\mathbb{R})^+$ acts on $\mathbb{H}$ by Mobius transformations and its center $Z(G(\mathbb{R})^+) = Z_\infty := \{zI : z \in \mathbb{R}^\times\}$ acts trivially.

Definition 2.11 Let $j_k : G(\mathbb{R})^+ \times \mathbb{H} \rightarrow \mathbb{C}$ where

$$j_k \left(g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, z \right) := (\det(g))^{-\frac{k}{2}} (cz + d)^{-k}. \tag{60}$$

Lemma 2.12 $\forall g_1, g_2 \in G(\mathbb{R})^+$ and $\forall z \in \mathbb{H}$,

$$j_k(g_1 g_2, z) \tag{61}$$

Proof $\forall \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \in G(\mathbb{R})^+$,

$$g_1 g_2 = \begin{pmatrix} * & * \\ c_1 a_2 + d_1 c_2 & c_1 b_2 + d_1 d_2 \end{pmatrix} \quad (62)$$

$$\begin{aligned} j_k(g_1, g_2 z) j_k(g_2, z) &= \left(c_1 \left(\frac{a_2 z + b_2}{c_2 z + d_2} \right) + d_1 \right)^{-k} (c_2 z + d_2)^{-k} \\ &= \left(c_1 (c_2 z + d_2) \left(\frac{a_2 z + b_2}{c_2 z + d_2} \right) + d_1 (c_2 z + d_2) \right)^{-k} \\ &= (c_1 (a_2 z + b_2) + d_1 (c_2 z + d_2))^{-k} \\ &= ((c_1 a_2 + d_1 c_2) z + (c_1 b_2 + d_1 d_2))^{-k} = j_k(g_1 g_2, z) \end{aligned} \quad (63)$$

□

$\forall g \in G(\mathbb{R})^+$ and $z \in \mathbb{R}^x$,

$$\begin{aligned} j(zg, \tau) &= (\det(zg))^{-\frac{k}{2}} (zcz + zd)^{-k} \\ &= z^{-k} \det(g)^{-\frac{k}{2}} z^k (cz + d)^{-k} \\ &= \frac{z}{|z|} j(g, z) \end{aligned} \quad (64)$$

Furthermore, $k_\theta = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \in \text{SO}_2(\mathbb{R}) =: K_\infty^+ \subset G(\mathbb{R})^+$, the maximal compact subgroup of $G(\mathbb{R})^+$.

Lemma 2.13 $Z_\infty K_\infty^+$ is the stabilizer of i .

Proof Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(\mathbb{R})^+$ stabilize i . Thus

$$\begin{aligned} \frac{ai + b}{ci + d} = i &\Rightarrow ai + b = di - c \\ &\Rightarrow a = d \wedge b = -c \end{aligned} \quad (65)$$

Thus

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix} \cdot i = i \quad (66)$$

. Let $k = \det \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Let

$$\begin{pmatrix} q & r \\ -r & q \end{pmatrix} = \frac{1}{k} \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \Rightarrow \begin{pmatrix} q & r \\ -r & q \end{pmatrix} \in \text{SL}_2(\mathbb{R}) \quad (67)$$

$\begin{pmatrix} q & r \\ -r & q \end{pmatrix} \in \text{SO}_2(\mathbb{R})$, by definition. Thus $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{k} I \begin{pmatrix} q & r \\ -r & q \end{pmatrix} \in Z_\infty K_\infty^+$. Thus $\text{Stab}_{G(\mathbb{R})^+}(i) \subset Z_\infty K_\infty^+$.

$\forall \lambda I \in Z_\infty^+$ and $\forall k_\theta \in K_\infty^+$,

$$\begin{aligned} (\lambda I k_\theta) i &= \frac{\cos(\theta)i + \sin(\theta)}{-\sin(\theta)i + \cos(\theta)} \frac{\cos(\theta) + i \sin(\theta)}{\cos(\theta) + i \sin(\theta)} \\ &= \frac{\cos(\theta)^2 i - \cos(\theta) \sin(\theta) + \cos(\theta) \sin(\theta) + i \sin(\theta)}{1} \\ &= (\cos(\theta)^2 + \sin(\theta)^2) i = i \end{aligned} \quad (68)$$

Thus $\text{Stab}_{G(\mathbb{R})^+}(i) = Z_\infty K_\infty^+$.

□

Additionally,

$$j_k(\lambda k_\theta, i) = \left(\frac{\lambda}{|\lambda|} \right)^{-k} (-\sin(\theta)i + \cos(\theta))^{-k} = \frac{\lambda}{|\lambda|} e^{ik\theta} \quad (69)$$

Definition 2.14 For $f \in S_k(\Gamma_0(N), \chi_N)$, let $\varphi = \varphi_f := G(\mathbb{R})^+ \rightarrow \mathbb{C}$ where

$$\varphi(g) = \varphi_f(g) = j_k(g, i) f(gi) \quad (70)$$

Lemma 2.15

1. For $\gamma \in \Gamma_0(N)$ and $g \in G(\mathbb{R})^+$,

$$\varphi(\gamma g) = \varphi(g) \chi_N(d) \quad (71)$$

2. $\forall \lambda k_\theta \in Z_\infty K_\infty^+$ and $\forall g \in G(\mathbb{R})^+$

$$\varphi(g \lambda k_\theta) = \left(\frac{\lambda}{|\lambda|} \right)^{-k} \varphi(g) e^{ik\theta} \quad (72)$$

Proof

1. $\varphi(\gamma g) = j_k(\gamma g, i) f(\gamma gi) = j_k(\gamma g, i) j_{-k}(\gamma, gi) \chi_N(d) f(gi)$. By Lemma 2.12,

$$\varphi(\gamma g) = j_k(\gamma, gi) j_k(g, i) j_{-k}(\gamma, gi) \chi_N(d) f(gi) = \chi_N(d) \varphi(g) \quad (73)$$

2.

$$\begin{aligned} \varphi(g \lambda k_\theta) &= j_k(g \lambda k_\theta, i) f(g \lambda k_\theta i) \\ &= j_k(g \lambda k_\theta, i) f(gi) \\ &= j_k(g, \lambda k_\theta i) j_k(\lambda k_\theta, i) f(gi) \\ &= \varphi(g) \left(\frac{\lambda}{|\lambda|} \right)^{-k} e^{ik\theta} \end{aligned} \quad (74)$$

□

You can extend φ to a function on $G(\mathbb{A})$ using the coset decompositon where $H = K_0(N)$ by setting

$$\varphi(g) = \varphi(\gamma g_\infty k) = \varphi(g_\infty) \chi(k). \quad (75)$$

This was all done without considering the fact that f is holomorphic.

Lemma 2.16 A smooth function $f : \mathbb{H} \rightarrow \mathbb{C}$ with the modularity property of $\Gamma_0(N)$ and χ_N is holomorphic if and only if φ_f satisfies $X_- \varphi = 0$ (Lie algebra action) where

$$X_\pm = \frac{1}{2} \begin{pmatrix} 1 & \pm i \\ \pm i & -1 \end{pmatrix} \quad (76)$$

Proof The real lie algebra

$$\mathfrak{g}_0 = \text{Lie}(G(\mathbb{R})) = M_2(\mathbb{R}) \quad (77)$$

acts on smooth functions on $G(\mathbb{R})^+$ by

$$X\psi(g_\infty) = \frac{d}{dt} \{ \psi(g_\infty \exp(tX)) \} |_{t=0}. \quad (78)$$

For example $X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, then

$$\begin{aligned}
\exp(tX) &= \sum_{n=0}^{\infty} \frac{1}{n!} t^n \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^n \\
&= \sum_{n=0}^{\infty} \left(\frac{t^{4n}}{(4n)!} I + \frac{t^{4n+1}}{(4n+1)!} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \frac{t^{4n+2}}{(4n+2)!} I - \frac{t^{4n+3}}{(4n+3)!} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right) \\
&= I \sum_{n=0}^{\infty} \frac{t^{2n}(-1)^n}{(2n)!} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \sum_{n=0}^{\infty} \frac{t^{2n+1}}{(2n+1)!} (-1)^n \\
&= \cos(t)I + \sin(t) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = k_t \in K_{\infty}^+
\end{aligned} \tag{79}$$

By Lemma 2.15,

$$\varphi(g_{\infty} \exp(tX)) = \varphi(g_{\infty} k_t) = \varphi(g_{\infty})(e^{it})^k \tag{80}$$

Thus

$$X\varphi(g_{\infty}) = \frac{d}{dt} \left\{ \varphi(g_{\infty})(e^{it})^k \right\} \Big|_{t=0} = \varphi(g_{\infty}) \frac{d}{dt} \{ e^{ikt} \} \Big|_{t=0} = ik\varphi(g_{\infty}). \tag{81}$$

This action of $\mathfrak{g}_0$ can be extended to $\mathfrak{g} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$ by setting $(X + iY)\varphi = X\varphi + iY\varphi$.

Let $X_{\pm}$ be defined as above. By checking the matrix multiplication we see

$$k_{\theta} X_{\pm} k_{\theta}^{-1} = e^{\pm 2i\theta} X_{\pm} \tag{82}$$

$$\begin{aligned}
X_{\pm} \varphi(g_{\infty} k_{\theta}) &= \frac{d}{dt} \{ \varphi(g_{\infty} k_{\theta} \exp(tX_{\pm})) \} \Big|_{t=1} \\
&= \frac{d}{dt} \{ \varphi(g_{\infty} \exp(tk_{\theta} X_{\pm} k_{\theta}^{-1} k_{\theta})) \} \Big|_{t=0} \\
&= \frac{d}{dt} \{ \varphi(g_{\infty} \exp(tk_{\theta} X_{\pm} k_{\theta}^{-1})) k_{\theta} \} \Big|_{t=0} \\
&= \frac{d}{dt} \{ \varphi(g_{\infty} \exp(te^{\pm 2i\theta} X_{\pm})) k_{\theta} \} \Big|_{t=0} \\
&= \frac{d}{dt} \{ \varphi(g_{\infty} \exp(te^{\pm 2i\theta} X_{\pm})) e^{ik\theta} \} \Big|_{t=0}
\end{aligned} \tag{83}$$

Last equality comes from Lemma 2.15. Let $s = te^{\pm 2i\theta} \Rightarrow \frac{d}{dt} = \frac{ds}{dt} \frac{d}{ds} = e^{\pm 2i\theta} \frac{d}{ds}$

Thus,

$$X_{\pm} \varphi(g_{\infty} k_{\theta}) = e^{ik\theta} e^{\pm 2i\theta} \frac{d}{ds} \{ \varphi(g_{\infty} \exp(sX_{\pm})) \} \Big|_{s=0} = e^{i\theta(k \pm 2)} X_{\pm} \varphi(g_{\infty}) \tag{84}$$

Thus X_{+} increases the weight by 2 and X_{-} decreases the weight by 2. Thus for the representations of GL_2 corresponding to holomorphic representations, $X_{-}\varphi = 0$ is the lowest weight vector condition and is equivalent to holomorphicity.

Let $z = x + iy$. Let

$$g_z = \begin{pmatrix} y^{\frac{1}{2}} & xy^{-\frac{1}{2}} \\ 0 & y^{-\frac{1}{2}} \end{pmatrix}. \tag{85}$$

$$g_z \cdot i = \frac{y^{\frac{1}{2}}i + xy^{-\frac{1}{2}}}{y^{-\frac{1}{2}}} = yi + x = z \text{ and } j_k(g_z, i) = y^{\frac{k}{2}} \quad (86)$$

Let $X_- = A + Bi$ where $A = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $B = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

$$\begin{aligned} (A\varphi)(g_z) &= \frac{d}{dt} \{ \varphi(g_z \exp(tA)) \} \big|_{t=0} \\ &= \frac{d}{dt} \{ j_k(g_z \exp(tA), i) f(g_z \exp(tA) \cdot i) \} \big|_{t=0} \\ (B\varphi)(g_z) &= \frac{d}{dt} \{ \varphi(g_z \exp(tB)) \} \big|_{t=0} \\ &= \frac{d}{dt} \{ j_k(g_z \exp(tB), i) f(g_z \exp(tB) \cdot i) \} \big|_{t=0} \end{aligned} \quad (87)$$

$$\begin{aligned} g_z \exp(tA) &= g_z \begin{pmatrix} e^{\frac{t}{2}} & 0 \\ 0 & e^{-\frac{t}{2}} \end{pmatrix} \\ &= \begin{pmatrix} y^{\frac{1}{2}} & xy^{-\frac{1}{2}} \\ 0 & y^{-\frac{1}{2}} \end{pmatrix} \begin{pmatrix} e^{\frac{t}{2}} & 0 \\ 0 & e^{-\frac{t}{2}} \end{pmatrix} \\ &= \begin{pmatrix} (e^t y)^{\frac{1}{2}} & x(e^t y)^{-\frac{1}{2}} \\ 0 & (e^t y)^{-\frac{1}{2}} \end{pmatrix} \end{aligned} \quad (88)$$

and

$$\begin{aligned} g_z \exp(tB) &= g_z \left(\sum_{n=0}^{\infty} \frac{1}{n!} t^n B^n \right) \\ &= g_z \left(\sum_{n=0}^{\infty} \left(\frac{1}{2n!} t^{2n} I + \frac{1}{(2n+1)!} t^{2n+1} B \right) \right) \\ &= g_z (I \cosh t + B \sinh t) \\ &= \begin{pmatrix} y^{\frac{1}{2}} & xy^{-\frac{1}{2}} \\ 0 & y^{-\frac{1}{2}} \end{pmatrix} \begin{pmatrix} \cosh(t) & -\sinh(t) \\ -\sinh(t) & \cosh(t) \end{pmatrix} \\ &= \begin{pmatrix} y^{\frac{1}{2}} \cosh(t) - xy^{-\frac{1}{2}} \sinh(t) & xy^{-\frac{1}{2}} \cosh(t) - y^{\frac{1}{2}} \sinh(t) \\ -y^{-\frac{1}{2}} \sinh(t) & y^{-\frac{1}{2}} \cosh(t) \end{pmatrix} \end{aligned} \quad (89)$$

Thus,

$$\begin{aligned} j_k(g_z \exp(tA), i) f(g_z \exp(tA) \cdot i) &= y^{\frac{k}{2}} e^{\frac{kt}{2}} f \left(\frac{e^{\frac{t}{2}} y^{\frac{1}{2}} i + x e^{-\frac{t}{2}} y^{-\frac{1}{2}}}{e^{-\frac{t}{2}} y^{-\frac{1}{2}}} \right) \\ &= y^{\frac{k}{2}} e^{\frac{kt}{2}} f(e^t yi + x) = y^{\frac{k}{2}} e^{\frac{kt}{2}} f(x + iye^t) \end{aligned} \quad (90)$$

and

$$j_k(g_z \exp(tB), i) f(g_z \exp(tB) \cdot i) = y^{\frac{k}{2}} (\cosh(t) + i \sinh(t))^{-k} f \left(\frac{xy^{-\frac{1}{2}} \cosh(t) - y^{\frac{1}{2}} \sinh(t) + i(y^{\frac{1}{2}} \cosh(t) - xy^{-\frac{1}{2}} \sinh(t))}{y^{-\frac{1}{2}} \cosh(t) - iy^{-\frac{1}{2}} \sinh(t)} \right) \quad (91)$$

Thus,

$$\begin{aligned}
(A\varphi)(g_z) &= \frac{d}{dt} \left\{ y^{\frac{k}{2}} e^{\frac{kt}{2}} f(x + iye^t) \right\} \Big|_{t=0} \\
&= y^{\frac{k}{2}} \left[\frac{k}{2} e^{\frac{kt}{2}} f(x + iye^t) + e^{\frac{kt}{2}} \frac{d}{dt} \{ f(x + iye^t) \} \right]_{t=0} \\
&= y^{\frac{k}{2}} \left[\frac{k}{2} e^{\frac{kt}{2}} f(x + iye^t) + e^{\frac{kt}{2}} \left(\frac{\partial f}{\partial y} \frac{d(ye^t)}{dt} \right) \right]_{t=0} \\
&= y^{\frac{k}{2}} \left[\frac{k}{2} e^{\frac{kt}{2}} f(x + iye^t) + e^{\frac{kt}{2}} ye^t \left(\frac{\partial f}{\partial y} \right) \right]_{t=0} \\
&= y^{\frac{k}{2}} \left[\frac{k}{2} f(z) + y \left(\frac{\partial f}{\partial y} \right) (z) \right]
\end{aligned} \tag{92}$$

and after a tedious calculation you get,

$$(B\varphi)(g_z) = y^{\frac{k}{2}} \left[-y \frac{\partial f}{\partial x}(z) + \frac{ik}{2} f(z) \right] \tag{93}$$

Finally,

$$\begin{aligned}
(X_- \varphi)(g_z) &= \left[y^{\frac{k}{2}} \left(\frac{k}{2} f(z) + y \left(\frac{\partial f}{\partial y} \right) (z) \right) \right] + i \left[y^{\frac{k}{2}} \left[-y \frac{\partial f}{\partial x}(z) + \frac{ik}{2} f(z) \right] \right] \\
&= y^{\frac{k}{2}} \left(\frac{k}{2} f(z) - \frac{k}{2} f(z) + y \left(\frac{\partial f}{\partial y} - i \frac{\partial f}{\partial x} \right) (z) \right) \\
&= y^{\frac{k}{2}+1} \left(\frac{\partial f}{\partial y} - i \frac{\partial f}{\partial x} \right) (z) \\
&= y^{\frac{k}{2}+1} (-i) \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) (z) = -2iy^{\frac{k}{2}+1} \frac{\partial f}{\partial \bar{z}}(z)
\end{aligned} \tag{94}$$

Thus we can prove the statement.

Suppose $(X_- \varphi) = 0 \Rightarrow (X_- \varphi)(g_z) = 0$ for all $z \in \mathbb{H}$. The identity derived above implies that $\frac{\partial f}{\partial \bar{z}}(z) = 0 \Rightarrow f$ is holomorphic.

Suppose f is holomorphic. Then by the above $(X_- \varphi)(g_z) = 0$ for all $z \in \mathbb{H}$ and thus for all upper triangular matrices. Since you can decompose all $G(\mathbb{R})$ as $K_\infty U$ and by identity above, $X_- \varphi = 0$. $\square$

Definition 2.17 Let $\mathcal{A}_0 := \mathcal{A}_0(\text{hol}, k, N, \chi)$ be the space of functions φ on $G(A)$ with the following properties

1. $\forall \gamma \in G(\mathbb{Q})$ and $k \in K_0(N)$,

$$\varphi(\gamma g k) = \varphi(g) \chi(k) \tag{95}$$

2. $\forall g_f \in G(\mathbb{A}_f)$ the function $g_\infty \rightarrow \varphi(g_\infty g_f)$ is smooth on $G(\mathbb{R})^+$ invariant under Z_∞^+ and satisfies

$$\varphi(gk_\theta) = e^{ik\theta} \varphi(g) \quad (96)$$

and

$$X_- \varphi = 0 \quad (97)$$

3. The function is cuspidal ie

$$\int_{\mathbb{Q}/\mathbb{A}} \varphi\left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} g\right) dx = 0 \quad (98)$$

for all $g \in G(\mathbb{A})$

Theorem 2.18 The map $f \rightarrow \varphi_f$ determines a isomorphism

$$S_k(\Gamma_0(N), \chi) \cong \mathcal{A}_0(\text{hol}, k, N, \chi) \quad (99)$$

Just for context, $\forall \varphi \in \mathcal{A}_0$, $\forall \gamma \in G(\mathbb{Q})$ and $\forall z \in \mathbb{Z}_\infty^+$ and $\forall g \in G(\mathbb{A})$, by previous lemmas we know:

$$\varphi(\gamma g) = \varphi(g) \text{ and } \varphi(zg) = \varphi(g) \quad (100)$$

Thus, φ factors through $(G(\mathbb{Q})Z_\infty^+)/G(\mathbb{A})$.

2.1. The representation theory of $\mathcal{A}_0(G)$

We want to bring the representation theory of $G(\mathbb{A})$ into play. But to use representation theory we need a space that is stable under the action of the group. The space $\mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ and the space of classical modular forms is not stable under the action of the full adelic group $G(\mathbb{A})$.

What that means:

1. **Group action:** $G(\mathbb{A})$ acts on $V = \mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ where $(g \cdot \varphi)(h) = \varphi(hg)$.
2. **The Problem:** The space $\mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ has very strict conditions:
 - Specific invariance on the right by $K_0(N)$
 - A specific weight k property: $\varphi(gk_\theta) = \varphi(g)(e^{i\theta})^k$
 - Holomorphy condition of $X_- \varphi = 0$.

So if you take a function $\varphi \in V$ that satisfies this and act on it by $h \in G(\mathbb{A})$. $(h \cdot \varphi)$ will not satisfy this condition. It will be invariant under a different subgroup $(h^{-1}K_0(N)h)$ and its weight and holomorphy properties will be mixed up.

3. We need to enlarge our space of functions to $\mathcal{A}_0(G)$ that we define to be stable under the group action.

So how do we enlarge our space of functions.

1. **Naive Idea:** The most obvious idea is to take φ and throw in $\text{span}_{\mathbb{C}}\{g \cdot \varphi : g \in G(\mathbb{A})\}$.
2. **Problem :** The issue is that φ , which has a nice pure weight k , when you translate it by $g \in G(\mathbb{R}) - \text{SO}_2(\mathbb{R})$ the new function $g \cdot \varphi$ becomes a ugly, possibly infinite, combination of different weights and thus our space is not stable under the action of $G(\mathbb{R})$
3. **Solution** Since the group $G(\mathbb{R})$ doesn't work nicely, we will use its Infinitesimal version: the Lie algebra $\mathfrak{g}$. The action of $\mathfrak{g}$ is essentially an action by differential operators. It turns out that the space of functions that are "finite sums of pure weights" is stable under the action of the Lie algebra $\mathfrak{g}$.

Let ω be a character of $\mathbb{Q}^\times/\mathbb{A}^\times$ with ω_∞ trivial on $\mathbb{R}_+^\times$ <https://chat.qwen.ai/s/f1b2d322-a25c-408a-bbf0-c454ec68fd66?fev=0.0.233>

Definition 2.1.1 The space of **automorphic forms** $\mathcal{A}(G) = \mathcal{A}(G, \omega)$ on $G(\mathbb{A})$ with central character ω is the space of $\mathbb{C}$ -valued functions φ on $G(\mathbb{A})$ such that:

1. For $z \in Z(\mathbb{A}) \cong \mathbb{A}^\times$ and $\gamma \in G(\mathbb{Q})$, $\varphi(z\gamma g) = \omega(z)\varphi(g)$ (Functions factor through $(G(\mathbb{Q})Z_\infty^+)/G(\mathbb{A})$) also φ is an eigenfunction under the action of $Z(\mathbb{A})$.
2. For each $\gamma \in G(\mathbb{A}_f)$ the function on $G(\mathbb{R})^+$, $g \rightarrow \varphi(g\gamma)$ is smooth.
3. The space spanned by right translates of φ by elements of K_∞ is finite dimensional, i.e. φ is right K_∞ finite. Thus a function φ is a finite sum of different pure weight functions?
4. There is a compact open subgroup K' of $G(\mathbb{A}_f)$ such that φ is invariant under right translation by K' . We may as well assume that $K' \subset K$ (definition of K in Equation 36). Examples include K , $K_0(N)$, etc.
5. $Z(\mathfrak{g})$ -finite: $Z(\mathfrak{g})$ is the center of the universal enveloping algebra where $\mathfrak{g} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$. Elements of $Z(\mathfrak{g})$ act as differential operators that commute with the action of $G(\mathbb{R})$ and the most important one is the Casimir (It returned!). $Z(\mathfrak{g})$ finiteness that φ lies in the generalized eigenspace for all Casimir type operators. This condition ensures the representation has bounded infinitesimal character, which is crucial for
 - Preventing pathological growth
 - Ensuring the representation decomposes nicely
 - Connecting to spectral theory

Note the casamire is

$$C = H^2 + 2X_+X_- + 2X_-X_+ \quad (101)$$

where $H = -i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This condition tells you that there is a polynomial $R[T] \in C[T]$ such that $R(C)\varphi = 0$.

1. The function φ is slowly increasing: We need the growth condition to ensure convergence of integrals.

For the growth condition, we use the norm on $G(\mathbb{A})$. There is a embedding

$$G(\mathbb{A}) \hookrightarrow M_2(\mathbb{A}) \times \mathbb{A} \cong \mathbb{A}^5 \text{ where } g \rightarrow (g, \det(g)^{-1}). \quad (102)$$

For each place v , and $x = (x_i) \in \mathbb{A}^5$ put

$$|x|_v = \max_i (|x_i|_v) \quad (103)$$

and let $|x| = \prod_v |x|_v$. Note for example, that if $v \neq \infty$ and if $g_v \in K_v$ then $\det(g_v)$ is a unit so $|g|_v = 1$. Thus almost all factors in the product are 1. Then φ is slowly increasing or of moderate growth if there is an integer n such that

$$|\varphi(g)| = O(|g|^n) \quad (104)$$

Definition 2.1.2 The space $\mathcal{A}_0(G) = \mathcal{A}_0(G, \omega) \subset \mathcal{A}(G)$ is the space of cuspidal automorphic forms with central character ω and is defined by adding the cuspidal condition that φ is cuspidal if

$$\int_{\mathbb{Q}/\mathbb{A}} \varphi \left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} g \right) dx = 0 \quad (105)$$

for all $g \in G(\mathbb{A})$.

It is easy to see that $\mathcal{A}_0(G)(\text{hol}, k, N, \chi) \subset \mathcal{A}_0(G)$. The weight and holomorphy condition in $\mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ imply that

$$C\varphi = k(k-2)\varphi \quad (106)$$

for $\varphi \in \mathcal{A}_0(G)(\text{hol}, k, N, \chi)$.

Definition 2.1.3 Let $G(\mathbb{R})$ acts on $\mathfrak{g}$ by conjugation and call the action Adj .

Definition 2.1.4

- A $(\mathfrak{g}, K_\infty)$ –module is a $\mathbb{C}$ -representation V of $\mathfrak{g}$ and K_∞ such that the two actions are compatible ie

$$k \cdot X \cdot v = \text{Adj}(k)(X) \cdot k \cdot v \quad (107)$$

and for $X \in \text{Lie}(K_\infty) \subset \mathfrak{g}$

$$X \cdot v = \frac{d}{dt}(\exp(tX) \cdot v) \big|_{t=0} \quad (108)$$

- A $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ is a $(\mathfrak{g}, K_\infty)$ module with a smooth action of $G(\mathbb{A}_f)$ commuting with the action of $(\mathfrak{g}, K_\infty)$. Here smooth means that $v \in V$ is fixed by some compact open subgroup $K' \subset G(\mathbb{A}_f)$.

Thus $\mathcal{A}(G)$ and $\mathcal{A}_0(G)$ are $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ modules

Definition 2.1.5

1. A $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ module (π, V) is admissible if, for every irreducible representation σ of $K_\infty \times K$, the multiplicity of σ in V is finite.
2. (π, V) is irreducible if it has no proper subspaces preserved by $\mathfrak{g}, K_\infty$, and $G(\mathbb{A}_f)$.

There are analogues for this when we look at the local definitions. The case for nonarchimedean v is standard. For archimedean place v , a representation V of G_v is smooth if for every $w \in V$ is fixed by open subgroup and admissible if and only if for every open subgroup K , the space V^K is finite dimensional, and irreducible if it has no proper subspaces. Furthermore an irreducible admissible representation (π, V) is unramified or spherical if the space of K_v invariants is nonzero. In this case $\dim V^{K_v} = 1$.

Theorem 2.1.6 Suppose that, for each archimedean (respectively nonarchimedean) place v of $\mathbb{Q}$ (π_v, V_v) is an irreducible, admissible $(\mathfrak{g}_v, K_v)$ -module (respectively representation of G_v). Suppose, moreover, that for almost all $p \in \mathbb{P}$, the representation (π_p, V_p) is unramified and a basis vector $\xi_v^0 \in V_p^{K_p}$ is given. Then the restricted tensor product,

$$V = V_\infty \otimes \bigotimes_{p \in \mathbb{P}}^* V_p \quad (109)$$

with respect to the vectors ξ_v^0 is a irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ module.

Conversely if (π, V) is a irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -module then there is a collection $\{(\pi_v, V_v)\}_v$ such that $V \cong V_\infty \otimes_{p \in \mathbb{P}}^* V_p$.

Proof $\Rightarrow$: We will define the $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ module structure for V . For $g = g_\infty g_f \in G(\mathbb{R})^+ \times G(\mathbb{A}_f)$,

$$\pi(g) \left(v_\infty \otimes \bigotimes_{p \in \mathbb{P}}^* v_p \right) = \pi_\infty(g_\infty) v_\infty \otimes \bigotimes_{p \in \mathbb{P}}^* \pi_p(g_p) v_p \quad (110)$$

since almost all $g_p \in K_p$ and almost all $v_p = \xi_p^0$, only finitely many factors change. For $X \in \mathfrak{g}$,

$$\pi(X) \left(v_\infty \otimes \bigotimes_{p \in \mathbb{P}}^* v_p \right) = \pi_\infty(X) v_\infty \otimes \bigotimes_{p \in \mathbb{P}}^* v_p \quad (111)$$

1. **Admissible:** To show this we have to show for every irreducible representation σ of $K_\infty \times K$ that $\dim V^\sigma < \infty$. Where $V^\sigma = \{v \in V \mid \pi(k)v = \sigma(k)v, \forall k \in K_\infty \times K\}$. We can write

$$\sigma = \sigma_\infty \times \bigotimes_{p \in \mathbb{P}}^* \sigma_p \quad (112)$$

where σ_p is trivial for almost all p since K is compact open subgroup. Thus $V^\sigma = V_\infty^{\sigma_\infty} \otimes \bigotimes_{p \in \mathbb{P}}^* V_p^{\sigma_p}$. V_∞ is a admissible representation, $\dim V_\infty^{\sigma_\infty} < \infty$ and same for $\dim V_p^{\sigma_p} < \infty$. Furthermore for almost all p , σ_p is trivial thus $V_p^{\sigma_p} = V_p^{K_p} = \mathbb{C}\xi_p^0$ which is 1 dimensional. Thus the tensor product is finite dimensional.

2. **Irreducible:** Suppose $W \subset V$ is a nonzero $(g, K_\infty) \times G(\mathbb{A}_f)$ -invariant subspace. We need to show that $W = V$. By Flath's Tensor Product Theorem, if π_v is irreducible for all v then $V = \bigotimes_v^* V_v$ is a irrep as a $(g, K_\infty) \times G(\mathbb{A}_f)$ module. The key part is that you can separate the variables. $\forall w \in W$ nonzero, $w = \sum_{i=1}^n w_\infty^i \otimes w_f^i$. By acting with $G(\mathbb{A}_f)$ and using irreducibility of π_p we can generate any vector in $\bigotimes_{p \in \mathbb{P}}^* V_p$. Similarly, by acting by $\mathfrak{g}$ and K_∞ you can get anything in V_∞ . Thus $W = \{v_\infty \otimes v_f : v_\infty \in V_\infty \wedge v_f \in V_f\} = V$.

$\Leftarrow$: For each prime p consider $V_p = \{v \in V : \exists K_p' \subset G(\mathbb{Q}_p) \text{ with } \pi(k)v = v, \forall k \in K_p'\}$. This is the space of K_p spooth vectors at p . $\square$

Theorem 2.1.7

1. The space $\mathcal{A}_0(G) = \mathcal{A}_0(G, \omega)$ is a algebraic direct sum of irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -modules,

$$\mathcal{A}_0(G) = \bigoplus_{\pi} \mathcal{A}_0(\pi) \quad (113)$$

2. Multiplicity One: If π_1 and π_2 are irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ modules, then

$$\pi_1 \cong \pi_2 \Rightarrow \mathcal{A}_0(\pi_1) = \mathcal{A}_0(\pi_2). \quad (114)$$

Thus $\dim \mathcal{A}_0(\pi) \leq 1$.

3. Strong Multiplicity One: Suppose that $\pi_{1,v} \cong \pi_{2,v}$ for all v outside of places of $\mathbb{Q}$ with density less than $\frac{1}{8}$. Then $\pi_1 \cong \pi_2$ and $\mathcal{A}_0(\pi) = \mathcal{A}_0(\pi_2)$

2.2. Local Representation Theory

The following is completely generalizable to a F be a p -adic field and $\mathcal{O}$ be the ring of integers, but we will use $\mathbb{Q}_p$ and $\mathbb{Z}_p$ for simplicity.

$$G = \mathrm{GL}_2(\mathbb{Q}_p) \text{ and } K = \mathrm{GL}_2(\mathbb{Z}_p) \quad (115)$$

so K is the maximal compact, and open, subgroup of G . Let B be the Borel subgroup.

$$B = \{\text{upper triangular}\} = \langle M, N \rangle \quad (116)$$

where $M = \{\text{diagonal matrices}\} = \{m(a) = \mathrm{diag}(a_1, a_2) \mid a = (a_1, a_2) \in \mathbb{Q}_p^\times \times \mathbb{Q}_p^\times\}$ (the maximally split torus in $\mathrm{GL}_2(\mathbb{Q}_p)$) and $N = \{n(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \mid x \in \mathbb{Q}_p\}$. $Z = Z(G)$. Let $\mu_1, \mu_2 : \mathbb{Q}_p^\times \rightarrow \mathbb{C}^\times$ be characters. Let $\mu : B \rightarrow \mathbb{C}^\times$ where $\begin{pmatrix} a & * \\ 0 & c \end{pmatrix} \rightarrow \mu_1(a)\mu_2(b)$.

$$\begin{aligned}
\mu\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} p & q \\ 0 & r \end{pmatrix}\right) &= \mu\left(\begin{pmatrix} ap & * \\ 0 & cr \end{pmatrix}\right) \\
&= \mu_1(ap)\mu_2(cr) \\
&= \mu_1(a)m_2(c)m_1(p)\mu_2(r) \\
&= \mu\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\right)\mu\left(\begin{pmatrix} p & q \\ 0 & r \end{pmatrix}\right)
\end{aligned} \tag{117}$$

Thus μ is a character of B . Let $\delta : \mathbb{B} \rightarrow \mathbb{R}_{>0}$ where

$$\delta\left(\begin{pmatrix} a_1 & * \\ 0 & a_2 \end{pmatrix}\right) = \left|\frac{a_1}{a_2}\right|_p \tag{118}$$

. Let

$$I(\mu) = \text{Ind}_G^B(\mu \cdot \delta^{\frac{1}{2}}) = \left\{ f : G \rightarrow \mathbb{C} : f(bg) = \mu(b)\delta(b)^{\frac{1}{2}}f(g) \right\} \tag{119}$$

where f is smooth, ie right invariant under an open subgroup $K' \subset G$ (this depends on f , and wlog $K' \subset K$).

$$d(bnb^{-1}) = \delta(b) dn \tag{120}$$

Thus we normalize the induction so that you get a unitarizable representation $I(\mu)$. The Iwasawa Decomposition tells us that $G = BK$. This gives us

Lemma 2.2.1

$$I(\mu) \cong I_{B \cap K}^K(\mu) := \text{Ind}_{B \cap K}^K(\mu \cdot \delta^{\frac{1}{2}}) \tag{121}$$

Proof $\varphi : I(\mu) \rightarrow I_{B \cap K}^K(\mu)$ where $f \rightarrow f|_K$.

1. Well Defined: $\forall h \in K \cap B$ and $k \in K$ we have

$$f(hk) = \mu(h)\delta^{\frac{1}{2}}(h)f(k) \tag{122}$$

2. Injective: Suppose $f|_K = 0$. Then for $g \in G(\mathbb{Q}_p)$, $g = bk$ by the iwasawa decomposition. $f(g) = \mu(b)\delta(b)^{\frac{1}{2}}f(k) = 0$.

3. Surjective: It is surjective by definition.

Thus φ is an isomorphism. □

Theorem 2.2.2

1. $I(\mu)$ is irreducible if and only if $\mu_1\mu_2^{-1} = \omega_{\pm 1}(x)$ where $\omega(x) = |x|_p^s$
2. If $\mu_1\mu_2^{-1} = \omega_1 \Rightarrow I(\mu)$ has a one dimensional quotient on which G acts by the character $\chi \circ \det$, where $\mu = (\chi\omega_{\frac{1}{2}}, \chi\omega_{-\frac{1}{2}})$ and an the infinite dimensional irreducible subrepresentation $\sigma(\mu)$
3. If $\mu_1\mu_2^{-1} = \omega_{-1} \Rightarrow I(\mu)$ has a one dimensional submodule on which G acts by the character $\chi \circ \det$ where $\mu = (\chi\omega_{-\frac{1}{2}}, \chi\omega_{\frac{1}{2}})$ and on the infinite dimensional quotient $\sigma(\mu)$
4. The only equivalences among these representations are teh following. Let $\nu = (\mu_2, \mu_1)$. If $\mu_1\mu_2^{-1} \neq \omega_{\pm 1}$ then $I(\mu) \cong I(\mu')$. If $\mu_1\mu_2^{-1} = \omega_{\pm 1}$ then $\sigma(\mu) \cong \sigma(\nu)$

Proof $N(M) = \left\langle M, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\rangle$. Thus $W = N(M) \setminus M = \left\langle w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\rangle \cong \mathbb{Z}_2$ is the weyl group.

Let $T : I(\mu = (\mu_1, \mu_2)) \rightarrow I(\nu = (\mu_2, \mu_1))$ where

$$f \rightarrow \left[g \rightarrow \int_N f(wng)dn = \int_{\mathbb{Q}_p} f(wn(x)g) dx \right] \quad (123)$$

$$\begin{aligned}
(Tf) \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} g \right) &= \int_{\mathbb{Q}_p} f \left(wn(x) \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} g \right) dx \\
&= \int_{\mathbb{Q}_p} f \left(wn(x) \begin{pmatrix} 1 & \frac{b}{c} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} g \right) dx \\
&= \int_{\mathbb{Q}_p} f \left(wn \left(x + \frac{b}{c} \right) \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} g \right) dx \\
&= \int_{\mathbb{Q}_p} f \left(wn(x) \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} g \right) dx \\
&= \int_{\mathbb{Q}_p} f \left(w \begin{pmatrix} a & xc \\ 0 & c \end{pmatrix} g \right) dx \\
&= \int_{\mathbb{Q}_p} f \left(w \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & \frac{xc}{a} \\ 0 & 1 \end{pmatrix} g \right) dx \\
&= \int_{\mathbb{Q}_p} f \left(w \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & \frac{xc}{a} \\ 0 & 1 \end{pmatrix} g \right) dx \\
&= \left| \frac{a}{c} \right|_p \int_{\mathbb{Q}_p} f(m(c, a)wn(x)g) dx
\end{aligned} \quad (124)$$

We can use the fact that $f \in I(\mu)$, thus

$$\begin{aligned}
(Tf) \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} g \right) &= \left| \frac{a}{c} \right|_p \int_{\mathbb{Q}_p} \mu_1(c) \mu_2(a) \left| \frac{c}{a} \right|_p^{\frac{1}{2}} f(wn(x)g) dx \\
&= \mu_2(a) \mu_1(c) \left| \frac{a}{c} \right|_p^{\frac{1}{2}} \int_{\mathbb{Q}_p} f(wn(x)g) dx \\
&= \nu \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) \delta \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right)^{\frac{1}{2}} (Tf)(g)
\end{aligned} \quad (125)$$

Thus $Tf \in I(\nu)$. Note but this is all depending on the convergence of Tf . Let f be a spherical function (invariant under K action on the right). By the iwasawa decomposition, $\forall g \in G(\mathbb{Q}_p)$, $g = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} k$ where $a, c \neq 0$ and $k \in K$.

By above and the fact that f_0 is a spherical vector,

$$\begin{aligned}
(Tf_0)(g) &= (Tf_0) \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} k \right) = \int_{\mathbb{Q}_p} f_0 \left(wn(x) \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} k \right) dx \\
&= \nu \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) \delta \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right)^{\frac{1}{2}} \int_{\mathbb{Q}_p} f_0(wn(x)k) dx \\
&= \nu \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) \delta \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right)^{\frac{1}{2}} \int_{\mathbb{Q}_p} f_0(wn(x)) dx
\end{aligned} \quad (126)$$

By the Iwasawa decomposition, $wn(x) = \beta_x k_x$ for $\beta_x \in B$ and $k_x \in K$.

1. Case $x \in \mathbb{Z}_p$: $wn(x) \in K$ and $\beta_x = I$.
2. Case $x \notin \mathbb{Z}_p$:

$$wn(x) = \begin{pmatrix} -\frac{1}{x} & 1 \\ 0 & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \frac{1}{x} & 1 \end{pmatrix} \quad (127)$$

Thus

$$\begin{aligned} \int_{\mathbb{Q}_p} f_0(wn(x)) dx &= \int_{\mathbb{Z}_p} f_0(wn(x)) dx + \int_{\mathbb{Q}_p - \mathbb{Z}_p} f_0(wn(x)) dx \\ &= \int_{\mathbb{Z}_p} f_0(I) dx + \int_{\mathbb{Q}_p - \mathbb{Z}_p} f_0 \left(\begin{pmatrix} -\frac{1}{x} & 1 \\ 0 & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \frac{1}{x} & 1 \end{pmatrix} \right) dx \\ &= \int_{\mathbb{Z}_p} 1 dx + \int_{\mathbb{Q}_p - \mathbb{Z}_p} \mu_2(x) \mu_1 \left(-\frac{1}{x} \right) \left| -\frac{1}{x^2} \right|_p^{\frac{1}{2}} f_0(I) dx \\ &= \int_{\mathbb{Z}_p} 1 dx + \mu_1(-1) \int_{\mathbb{Q}_p - \mathbb{Z}_p} (\mu_2 \mu_1^{-1})(x) |x|_p^{-1} dx \end{aligned} \quad (128)$$

□

Theorem 2.2.3 For an irreducible admissible representation (π, V) of G the following are equivalent:

1. For all μ , $\text{Hom}_G(\pi, I(\mu)) = 0$
2. The matrix coefficients $\varphi_{v,v'}$ of π are compactly supported moduler Z , ie, the image of $\frac{G}{Z}$ of the support of $\varphi_{v,v'}$. Note the space of matrix coefficients we will use $V \otimes \tilde{V}$ where $\tilde{V} \subset V^*$ of smooth vectors.

Such representations (π, V) are called **supercuspidal**.

The set of irreducible admissible representation of G are the following:

1. Supercuspidal representation
2. Irreducible principle series representations $I(\mu)$
3. The special Representations $\sigma(\mu)$
4. 1 dimensional representations $\sigma \circ \det$

A representation (π, V) of G is unramified if $\dim V^K > 0$ ie there exists a spherical vector. If the characters $\mu = (\mu_1, \mu_2)$ are unramified then

$$\mu_j(x) = t_j^{v_p(x)} \quad (129)$$

for some pair $(t_1, t_2) \in \mathbb{C}^\times \times \mathbb{C}^\times$ then $\mu|_{B \cap K} = 1$ and the restriction to K gives the isomorphism

$$I(\mu) \cong C^\infty(B \cap K/K) \quad (130)$$

this isomorphism is equivariant for the action of K by right multiplication. Thu $\dim I(\mu)^K = 1$.

Theorem 2.2.4

1. For every pair $(t_1, t_2) \in (\mathbb{C}^\times)^2$ there is a irreducible admissable unramified representation $\pi(t_1, t_2)$ of G .
2. Every irreducible admissible unramified representation of G is isomorphic to one of the $\pi(t_1, t_2)$.
The only equivalence between such representations is $\pi(t_2, t_1) \cong \pi(t_1, t_2)$
3. If $\mu_1 \mu_2^{-1} \neq \omega_{\pm 1}$ ie if $t_1 t_2^{-1} \neq q^{\pm 1}$ then $\pi(t_1, t_2) = I(\mu)$. Otherwise $\pi(t_1, t_2) = \chi \circ \det$ the one dimensional constituent of $I(\mu)$.

Remark 2.2.5 Let

$$I = \left\{ k = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K \mid v_p(c) \geq 1 \right\} = \left\{ k = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K \mid c \equiv 0 \pmod{p} \right\} \quad (131)$$

be the Iwahori Subgroup of K . Then the only irreducible admissable representations (π, V) of G with $V^I \neq 0$ are the unramified representations or the unramified special representations ie $\sigma(\mu)$ where μ is unramified.

Corollary 2.2.6 There is a isomorphism between the isomorphism classes of irreducible admissable unramified representations of G and the semisimple conjugacy classes of $G(\mathbb{C})$ given by

$$\pi = \pi(t_1, t_2) \leftrightarrow t(\pi) := \left[\begin{pmatrix} t_1 & \\ & t_2 \end{pmatrix} \right] \quad (132)$$

$t(\pi)$ is called the **Satake parameter** of π .

Definition 2.2.7 The (spherical) **Hecke Algebra**

$$\mathcal{H} = \mathcal{H}(G, K) = \mathbb{C}\langle K \backslash G / K \rangle \quad (133)$$

is the space of compactly supported functions ξ on G that are invariant under left and right multiplication by K . $\mathcal{H}$ is a algebra under convolution with the identity element being the characteristic function of K .

Let

$$A^+ := \left\{ \begin{pmatrix} p^{\lambda_1} & 0 \\ 0 & p^{\lambda_2} \end{pmatrix} : \lambda_1 \geq \lambda_2 \in \mathbb{Z} \right\}. \quad (134)$$

The Cartan Decomposition tells us that $G = K A^+ K$. Thus $K \backslash G / K = A$. We will look at two distinguished elements in A^+ , $\xi_1 = pI$ and $\xi_2 = \text{diag}(p, 1)$.

Thus $\{1_{K a K} : a = (a_1, a_2) = \text{diag}(p^{a_1}, p^{a_2}) \in A\}$ generate $\mathcal{H}$. For a unramified rep. V of G , $\mathcal{H}$ acts on V^K by

$$\xi(v) = \int_G \xi(g) g v \, dg = \lambda_V(\xi) v \quad (135)$$

Note $\dim V^K = 1$ (look above). Furthermore this means that $\lambda_V : \mathcal{H} \rightarrow \mathbb{C}$ is a algebra homomorphism.

Furthermore every algebra homomorphism $\mathcal{H} \rightarrow \mathbb{C}$ arises this way. In fact $\mathcal{H}$ can be viewed as the space of regular functions on the space of semisimple conjugacy classes of $G(\mathbb{C})$ via the map $\xi \rightarrow \lambda_\pi(\xi)$

Theorem 2.2.8

$$\begin{aligned}
\mathcal{H} &= \mathbb{C}[\xi_1, \xi_2] \cong \mathbb{C}[t_1 + t_2, t_1 t_2] \\
\xi_1 &\rightarrow t_1 t_2 = \lambda_\pi(\xi_2) \\
\xi_2 &\rightarrow p^{-\frac{1}{2}}(t_1 + t_2) = \lambda_\pi(\xi_1)
\end{aligned} \tag{136}$$

Proof By Corollary 2.2.6, $\left(\begin{smallmatrix} \mathbb{C}^\times \\ 2 \end{smallmatrix}\right)$ (conjugacy classes of $G(\mathbb{C})$) is identified with isomorphism classes of irreducible admissible unramified representations. Furthermore, $G = BK$. Thus $\forall t \in \left(\begin{smallmatrix} \mathbb{C}^\times \\ 2 \end{smallmatrix}\right)$, $(\pi(t), I(\mu))$ for $\mu\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\right) = \left(t_1^{\text{val}_p(a)}, t_2^{\text{val}_p(c)}\right)$. Let $\varphi \in I(\mu)^K$. For $\xi \in \mathcal{H}$,

$$\begin{aligned}
(\pi_t(\xi)\varphi)(e) &= \lambda_{\pi_t}(\xi)(\varphi)(e) = \lambda_{\pi_t}(\xi) \\
&= \int_G \xi(g)(\pi_t(g)\varphi)(e) \, dg \\
&= \int_G \xi(g)\varphi(ge) \, dg \\
&= \int_G \xi(g)\varphi(g) \, dg
\end{aligned} \tag{137}$$

Normalize the measure so $\text{vol}(K) = 1$.

Let $S : \mathcal{H} \rightarrow \text{Hom}\left(\left(\begin{smallmatrix} \mathbb{C}^\times \\ 2 \end{smallmatrix}\right), \mathbb{C}\right) = \mathbb{C}[t_1, t_2]^{\mathbb{Z}_2} = \mathbb{C}[t_1 + t_2, t_1 t_2]$ where

$$\xi \rightarrow [t = \{t_1, t_2\} \rightarrow \lambda_{\pi_t}(\xi)] \tag{138}$$

We want to show that S is a isomorphism. We will show that it is injective and that surjective just with ξ_1 and ξ_2 . Thus $\mathcal{H}$ is generated by ξ_1 and ξ_2 .

$$\begin{aligned}
\lambda_{\pi(t)}(1_{KaK}) &:= \int_G 1_{KaK}(g)\varphi(g) \, dg \\
&= \int_{KaK} \varphi(g) \, dg \\
&= \text{vol}(K) \int_K \varphi(ka) \, dk = \int_K \varphi(ka) \, dk
\end{aligned} \tag{139}$$

1. Calculation for $\xi_1 = KpIK = pIK = pK$:

$$\begin{aligned}
\lambda_{\pi(t)}(1_{pK}) &= \int_K \varphi(pk) \, dk \\
&= \int_K \varphi(pI) \, dk \\
&= \int_K t_1^1 t_2^1 p^0 \, dk = t_1 t_2
\end{aligned} \tag{140}$$

2. Calculation for $\xi_2 = 1_{K\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}K}$:

There is a coset decomposition of $K\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}K$ which is

$$K\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}K = \left(\sqcup_{j=0}^{p-1} c_j K\right) \sqcup c_\infty K \tag{141}$$

where

$$\begin{aligned}
c_j &= \begin{pmatrix} p & j \\ 0 & 1 \end{pmatrix} \\
c_\infty &= \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}.
\end{aligned} \tag{142}$$

Thus

$$\begin{aligned}
\lambda_{\pi_t}(\xi_2) &= \sum_{j=0}^{p-1} \int_K \varphi\left(\begin{pmatrix} p & j \\ 0 & 1 \end{pmatrix} k\right) \, dk + \int_K \varphi\left(\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} k\right) \, dk \\
&= \sum_{j=0}^{p-1} \int_K t_1^1 t_2^0 p^{-\frac{1}{2}} \, dk + \int_K t_1^0 t_2^1 p^{\frac{1}{2}} \, dk \\
&= \sum_{j=0}^{p-1} t_1^1 t_2^0 p^{-\frac{1}{2}} + t_1^0 t_2^1 p^{\frac{1}{2}} \\
&= p(t_1^1 t_2^0 p^{-\frac{1}{2}}) + t_1^0 t_2^1 p^{\frac{1}{2}} \\
&= p^{\frac{1}{2}}(t_1 + t_2)
\end{aligned} \tag{143}$$

$$\begin{aligned}
\lambda_{\pi(t)}(\alpha * \beta) &= \int_G \int_G \alpha(h) \beta(h^{-1}g) \pi(g) \, dh \, dg \\
&= \int_G \int_G \alpha(h) \beta(h^{-1}g) \pi(g) \, dg \, dh
\end{aligned} \tag{144}$$

Let $k = h^{-1}g$ since haar measure is left invariant, $dk = dg$. thus

$$\begin{aligned}
\lambda_{\pi(t)}(\alpha * \beta) &= \int_G \int_G \alpha(h) \beta(k) \pi(hk) dk dh \\
&= \int_G \int_G \alpha(h) \beta(k) \pi(h) \pi(k) dg dh \\
&= \left(\int_G \alpha(h) \pi(h) dh \right) \left(\int_G \beta(k) \pi(k) dk \right) \\
&= \lambda_{\pi_t}(\alpha) \lambda_{\pi_t}(\beta)
\end{aligned} \tag{145}$$

Thus by linearity the map is a homomorphism and surjective. All that is left to show is that $\ker S = 0$. Suppose $f \in \ker(S)$, then for all $t \in \left(\frac{\mathbb{C}}{2}\right)$

$$\lambda_{\pi_t}(f) = 0 = \int_G f(g) \pi(g) dg \tag{146}$$

□

2.3. Classical Newforms

Coming back to automorphic representations and the isomorphism:

$$\begin{aligned}
S_k(\Gamma(0)(N), \chi_N) &\xrightarrow{\cong} \mathcal{A}_0(G)(\text{hol}, k, N, \chi) \\
f &\rightarrow \varphi_f
\end{aligned} \tag{147}$$

Proposition 2.3.1 If p doesn't divide N , then φ_f is invariant under right translation by K_p .

Proof

$$K_0(N) = \prod_{p \in \mathbb{P}} K_0(p^{v_p(N)}) \tag{148}$$

Suppose $p \nmid N$, then $K_0(p^{v_p(N)}) = K_0(p^0) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : c = 0 \pmod{1} \right\} = G(\mathbb{Z}_p) = K_p$. Since $p \nmid N$, for $k_p \in K_p$, $\chi_p(k_p) = 1$. By the transformation law of φ_f we have

$$\varphi_f(g \cdot k_p) = \chi_p(k_p) \varphi_f(g) = \varphi_f(g) \tag{149}$$

□

Thus if $p \nmid N$, then $\mathcal{H}_p$ acts. On $S_k(\Gamma(0)(N), \chi_N)$, $\mathcal{H}_p \cong \mathbb{C}[\xi_{p,1}, \xi_{p,1}]$ act like standard classical hecke operators:

$$\begin{aligned}
\xi_{p,1} f_0 &= R_p f_0 \\
\xi_{p,2} f_0 &= p^{\frac{k-1}{2}} T_p f_0
\end{aligned} \tag{150}$$

Thus there is a action of $\mathcal{H}^N = \otimes_{p \nmid N} \mathcal{H}_p$

2.4. Rederiving Properties of Hecke Operators using Hecke Algebras

Remember $G_p := G(\mathbb{Q}_p)$ and $K_p := \mathbb{Z}_p$. Let $\mathcal{H} = \mathcal{H}(G_p, K_p)$. We want to study where $1_{K_p(n,1)K_p}$ maps to with the Satake isomorphism.

Theorem 2.4.1

Proof The coset decomposition of

$$K_p \begin{pmatrix} p^n & 0 \\ 0 & 1 \end{pmatrix} K_p := \bigsqcup_{i=0}^n \bigsqcup_{j=0}^{p^i-1} \begin{pmatrix} p^i & j \\ 0 & p^{n-i} \end{pmatrix} K_p \quad (151)$$

Thus

$$\begin{aligned} \lambda_{\pi(t)}(1_{K_p(p^n,1)K_p}) &= \sum_{i=0}^n \sum_{j=0}^{p^i-1} \int_K \varphi\left(\begin{pmatrix} p^i & j \\ 0 & p^{n-i} \end{pmatrix} k\right) dk \\ &= \sum_{i=0}^n \sum_{j=0}^{p^i-1} t_1^i t_2^{n-i} \left| \frac{p^i}{p^{n-i}} \right|^{\frac{1}{2}} \\ &= \sum_{i=0}^n \sum_{j=0}^{p^i-1} t_1^i t_2^{n-i} |p|^{\frac{-(n-2i)}{2}} \\ &= p^{\frac{n}{2}} \sum_{i=0}^n \sum_{j=0}^{p^i-1} t_1^i t_2^{n-i} p^{-i} \\ &= p^{\frac{n}{2}} \sum_{i=0}^n t_1^i t_2^{n-i} \end{aligned} \quad (152)$$

$$\begin{aligned} (t_1 + t_2) \sum_{i=0}^n t_1^i t_2^{n-i} &= \sum_{i=0}^n (t_1^{i+1} t_2^{n-i} + t_1^i t_2^{n-i+1}) \\ &= \sum_{i=0}^n (t_1^{i+1} t_2^{n-i} + t_1^i t_2^{n-i+1}) \\ &= \sum_{i=0}^{n+1} (t_1^i t_2^{(n+1)-i}) + (t_1 t_2) \sum_{i=0}^{n-1} t_1^i t_2^{n-1-i} \end{aligned} \quad (153)$$

Thus

$$\begin{aligned} 1_{K_p(p,1)K_p} * 1_{K_p(p^n,1)K_p} &\cong p^{\frac{n+1}{2}} (t_1 + t_2) \sum_{i=0}^n t_1^i t_2^{n-i} \\ &= p^{\frac{n+1}{2}} \left(\sum_{i=0}^{n+1} (t_1^i t_2^{(n+1)-i}) + (t_1 t_2) \sum_{i=0}^{n-1} t_1^i t_2^{n-1-i} \right) \\ &= p^{\frac{n+1}{2}} \sum_{i=0}^{n+1} (t_1^i t_2^{(n+1)-i}) + p^{\frac{n+1}{2}} (t_1 t_2) \sum_{i=0}^{n-1} t_1^i t_2^{n-1-i} \\ &= 1_{K_p(p^{n+1},1)K_p} + p 1_{(1_{K_p(p,p)K_p} * 1_{K_p(p^{n-1},1)K_p})} \end{aligned} \quad (154)$$

□

3. Appendix

Theorem 3.1 (*Cartan Decomposition*) Let $G = \mathrm{GL}_2(F)$ and $K = \mathrm{GL}_2(\mathcal{O})$. Then